Milestone_two

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```
library(bcdata)
```

```
Attaching package: 'bcdata'

The following object is masked from 'package:stats':

    filter
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1

-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks bcdata::filter(), stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
#bc_ses_data_raw <- bcdc_get_data("b-c-socioeconomic-index")
```

```
#write_csv(bc_ses_data_raw, "bc_ses_data_raw.csv")
```

After writing the dataset onto my computer, I have made a copy of the dataset to keep the original file.

```r
socioeconomic_index_raw <- read.csv("bc_ses_data_raw.csv")
```

Cleaning

```r
head(socioeconomic_index_raw)
```

```
  CSD_UID   MUN_NAME_2021 CALENDAR_YEAR AVERAGE_AGE POPULATION_ESTIMATE
1 5901003         Elkford          2023    37.34995                2949
2 5901006        Sparwood          2023    38.21653                4452
3 5901012          Fernie          2023    38.64001                6778
4 5901017 East Kootenay A          2023    45.75155                2003
5 5901019 East Kootenay B          2023    45.75155                2148
6 5901022       Cranbrook          2023    43.48266               22057
  TOTAL_INDEX_0_100 SEI_INDEX_0_100 ECON_0_100 EDUC_0_100 HEALTH_0_100
1          53.81362        51.90580   55.48008   44.59705     57.13935
2          51.14647        50.79946   52.92982   46.83267     51.41402
3          59.95708        59.24866   54.89918   60.41922     57.85204
4          56.03723        52.42306   49.75406   50.36227     63.34752
5          48.36915        47.10612   41.92556   49.19864     52.40895
6          45.34127        45.35226   47.96285   44.92382     47.26186
  COMMUNITY_0_100
1        54.85955
2        52.33694
3        57.14554
4        56.11092
5        52.17249
6        45.77451
```

First, I will get rid of columns that I do not need.

```r
cleaned_socioeconomic_index <- socioeconomic_index_raw |>
  select(-CALENDAR_YEAR)
```

Now, I will only include the municipalities of interest.

```r
cleaned_socioeconomic_index <- cleaned_socioeconomic_index |>
  filter(
    MUN_NAME_2021 %in% c(
```

```
    "Anmore", "Belcarra", "Bowen Island", "Burnaby",
    "Coquitlam", "Delta", "Langley", "Lions Bay", "Maple Ridge",
    "Metro Vancouver A", "New   Westminister", "North Vancouver",
    "Pitt Meadows", "Port Coquitlam", "Port Moody", "Richmond",
    "Surrey", "Vancouver", "West Vancouver", "White Rock", "Central Saanich",
    "Colwood", "Esquimalt", "Highlands", "Juan de Fuca (Part 1)",
    "Juan de Fuca (Part 2)", "Langford", "Metchosin", "North Saanich",
    "Oak Bay", "Saanich", "Sidney", "Sooke", "Victoria", "View Royal"
  )
)
```

In the dataset, some municipalities are named twice because they contain sub-regions or districts. It is important that we give these better identifiers so that we can be more precise in our analysis.

```
head(cleaned_socioeconomic_index)
```

```
  CSD_UID MUN_NAME_2021 AVERAGE_AGE POPULATION_ESTIMATE TOTAL_INDEX_0_100
1 5915001       Langley    39.72087              154103          56.11597
2 5915002       Langley    41.37304               33626          46.78980
3 5915004        Surrey    38.41333              653040          52.03304
4 5915007    White Rock    52.06709               24294          50.94842
5 5915011         Delta    41.76639              120048          54.09115
6 5915015      Richmond    42.06973              235082          54.86931
  SEI_INDEX_0_100 ECON_0_100 EDUC_0_100 HEALTH_0_100 COMMUNITY_0_100
1        56.71184   58.70496   52.74747     51.84272        54.85355
2        46.48592   51.75212   44.81775     49.05609        44.70281
3        51.49810   55.36133   46.14667     52.46490        52.31691
4        53.00977   51.16469   53.30810     44.76587        52.88104
5        54.78833   56.43488   50.12093     50.20275        55.22447
6        54.38329   48.56817   57.94051     54.28670        54.18359
```

```
cleaned_socioeconomic_index <- cleaned_socioeconomic_index |>
  mutate(MUN_NAME_2021 = case_when(
    CSD_UID == "5915002" ~ "City of Langley",
    CSD_UID == "5915001" ~ "Township of Langley",
    CSD_UID == "5915020" ~ "Electoral Area A (UBC)",
    CSD_UID == "5915051" ~ "City of North Vancouver",
    CSD_UID == "5915046" ~ "District of North Vancouver",
    CSD_UID == "5917054" ~ "Juan de Fuca (Part 1, Shirley, Jordan River, Malahat)",
    CSD_UID == "5917056" ~ "Juan de Fuca (Part 2, Port Renfrew)",
```

```
    TRUE ~ MUN_NAME_2021


  )
)

cleaned_socioeconomic_index
```

	CSD_UID	MUN_NAME_2021	AVERAGE_AGE
1	5915001	Township of Langley	39.72087
2	5915002	City of Langley	41.37304
3	5915004	Surrey	38.41333
4	5915007	White Rock	52.06709
5	5915011	Delta	41.76639
6	5915015	Richmond	42.06973
7	5915020	Electoral Area A (UBC)	37.80659
8	5915022	Vancouver	40.92182
9	5915025	Burnaby	40.73695
10	5915034	Coquitlam	40.26172
11	5915036	Belcarra	48.63754
12	5915038	Anmore	39.96462
13	5915039	Port Coquitlam	40.14932
14	5915043	Port Moody	39.65257
15	5915046	District of North Vancouver	41.74278
16	5915051	City of North Vancouver	41.55681
17	5915055	West Vancouver	46.43948
18	5915062	Bowen Island	45.20400
19	5915065	Lions Bay	45.32682
20	5915070	Pitt Meadows	41.15538
21	5915075	Maple Ridge	39.73551
22	5917005	North Saanich	50.31450
23	5917010	Sidney	55.16472
24	5917015	Central Saanich	45.91971
25	5917021	Saanich	43.54945
26	5917030	Oak Bay	49.01323
27	5917034	Victoria	44.21167
28	5917040	Esquimalt	43.25084
29	5917041	Colwood	41.58216
30	5917042	Metchosin	45.73052
31	5917044	Langford	38.45883
32	5917047	View Royal	43.05907
33	5917049	Highlands	42.56319

```
34 5917052                                                  Sooke    42.57043
35 5917054 Juan de Fuca (Part 1, Shirley, Jordan River, Malahat)    49.78713
36 5917056                   Juan de Fuca (Part 2, Port Renfrew)    49.78713
   POPULATION_ESTIMATE TOTAL_INDEX_0_100 SEI_INDEX_0_100 ECON_0_100 EDUC_0_100
1               154103          56.11597        56.71184   58.70496   52.74747
2                33626          46.78980        46.48592   51.75212   44.81775
3               653040          52.03304        51.49810   55.36133   46.14667
4                24294          50.94842        53.00977   51.16469   53.30810
5               120048          54.09115        54.78833   56.43488   50.12093
6               235082          54.86931        54.38329   48.56817   57.94051
7                21150          60.00909        57.08690   45.03683   66.64419
8               732593          55.74600        55.19496   54.27197   56.51232
9               283588          56.02838        55.38087   52.85356   56.28128
10              166588          56.02739        55.40461   54.57856   54.86436
11                 698          60.77746        61.08152   63.33843   53.46220
12                2600          63.58915        64.19955   65.43745   58.58725
13               66975          53.72884        53.79403   55.39012   51.09595
14               37677          59.90945        60.05184   61.08839   56.72302
15               96845          62.16632        63.33555   61.84869   62.86484
16               64735          57.35498        57.33259   55.65945   58.70918
17               47401          61.85727        64.02043   57.67481   70.13415
18                4505          60.28520        61.22227   57.40905   59.54638
19                1484          65.62385        66.45530   65.62139   63.95429
20               20929          51.15989        51.62288   56.19353   45.39978
21              102407          50.63409        50.71860   56.81419   43.86231
22               13224          58.40353        60.49946   59.61581   57.06146
23               13162          49.17739        52.28674   50.77390   49.58604
24               18358          55.32164        56.93065   57.93859   51.89960
25              125385          55.24898        56.01257   55.20089   55.01082
26               18892          60.90802        63.00385   59.17023   63.88717
27              100497          51.06384        51.22478   51.62005   54.14012
28               19016          48.17806        47.98201   51.54714   46.95490
29               21422          52.62275        52.76891   57.82651   46.26507
30                5299          54.43884        55.15177   54.64437   52.74902
31               55064          54.15588        53.37423   58.80071   47.00547
32               12578          54.42989        55.22583   57.07376   51.87593
33                2896          57.99720        57.84263   60.86034   51.44978
34               16796          47.71151        48.03726   53.01944   41.42974
35                5527          48.60174        48.01078   53.28835   43.57858
36                 430          47.05957        43.54016   33.71857   52.98479
   HEALTH_0_100 COMMUNITY_0_100
1      51.84272         54.85355
2      49.05609         44.70281
```

3	52.46490	52.31691
4	44.76587	52.88104
5	50.20275	55.22447
6	54.28670	54.18359
7	64.36135	55.47980
8	55.08025	51.80367
9	55.38549	53.91460
10	55.28437	53.65120
11	55.10600	60.30480
12	56.08570	60.47096
13	51.92042	52.73328
14	55.35830	56.53642
15	53.99040	57.47138
16	54.54784	53.25222
17	51.36929	55.83232
18	53.24463	60.45928
19	56.87245	60.22833
20	49.10556	52.48261
21	49.86988	51.15781
22	48.89412	58.91588
23	40.52026	55.50103
24	48.34298	57.17201
25	50.91676	54.38258
26	50.64094	58.40530
27	50.51568	46.96813
28	49.55654	46.44716
29	50.87479	52.72047
30	50.45364	55.23524
31	54.41595	52.41614
32	50.26889	53.74513
33	54.85572	56.84442
34	47.38607	51.01813
35	50.69349	48.31556
36	58.29669	47.75609